Solving Linear Systems: Iterative Methods

Direct vs. Iterative Methods

- So far, have looked at *direct methods* for solving linear systems
  - Predictable number of steps
  - No answer until the very end

- Alternative: *iterative methods*
  - Start with approximate answer
  - Each iteration improves accuracy
  - Stop once estimated error below tolerance

Numerical Solutions

Matrix form: $[\mathbf{A}][\mathbf{x}]=[\mathbf{b}]$.

From linear algebra: $[\mathbf{A}]^{-1}[\mathbf{A}][\mathbf{x}]=[\mathbf{A}]^{-1}[\mathbf{b}]$, $[\mathbf{x}]=[\mathbf{A}]^{-1}[\mathbf{b}]$

where $[\mathbf{A}]^{-1}$ is the inverse of matrix $[\mathbf{A}]$. $[\mathbf{x}]$ is the solution vector.

- Matrix inversion requires cumbersome numerical computations and is not efficient if the order of the matrix is high (>10)
- Gauss elimination method and other matrix solvers are usually available in many numerical solution package.
- For high order matrix, iterative methods are usually more efficient. The famous Jacobi & Gauss-Seidel iteration methods will be introduced in the following.

Benefits of Iterative Algorithms

- Some iterative algorithms designed for accuracy:
  - Direct methods subject to roundoff error
  - Iterate to reduce error to $O(\varepsilon)$
- Some algorithms produce answer faster
  - Speed depends on
    - No. of *nonzero* elements,
    - Not total of elements
- Today: iterative improvement of accuracy

Iterative Improvement

- Suppose we've to solve (or think we've solved) some system $Ax=b$
- Can check answer by computing *residual*:

$$r = b - Ax_{approximated}$$

- If r is small (compared to b), x is accurate
- What if it's not? diverging

Gauss-Seidel Method and Jacobi method,

Both are *iterative* method.

Basic Procedure:

- Algebraically solve each linear equation for x_i
- Assume an initial guess solution array
- Solve for each x_i and repeat
- Use absolute relative approximate error after each iteration to check if error is within a pre-specified tolerance.

Gauss-Seidel Method and Jacobi method,

Algorithm

A set of *n* equations and *n* unknowns:

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots + a_{2n}x_n = b_2$$

$$\vdots \qquad \vdots$$

$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \ldots + a_{nn}x_n = b_n$$

If: the diagonal elements are non-zero

Rewrite each equation solving for the corresponding unknown

ex: First equation, solve for x_1

Second equation, solve for x_2

Gauss-Seidel Method and Jacobi method,

Algorithm

Rewriting each equation

$$x_1 = \frac{b_1 - a_{12}x_2 - a_{13}x_3 \dots\dots - a_{1n}x_n}{a_{11}}$$ ← From Equation 1

$$x_2 = \frac{b_2 - a_{21}x_1 - a_{23}x_3 \dots\dots - a_{2n}x_n}{a_{22}}$$ ← From equation 2

$$\vdots \qquad \vdots \qquad \vdots$$

$$x_{n-1} = \frac{b_{n-1} - a_{n-1,1}x_1 - a_{n-1,2}x_2 \dots\dots - a_{n-1,n-2}x_{n-2} - a_{n-1,n}x_n}{a_{n-1,n-1}}$$ ← From equation n-1

$$x_n = \frac{b_n - a_{n1}x_1 - a_{n2}x_2 - \cdots \dots - a_{n,n-1}x_{n-1}}{a_{nn}}$$ ← From equation n

Gauss-Seidel Method and Jacobi method,

Algorithm

General Form of each equation

$$x_1 = \frac{b_1 - \sum_{\substack{j=1 \\ j\neq 1}}^{n} a_{1j} x_j}{a_{11}}$$

$$x_{n-1} = \frac{b_{n-1} - \sum_{\substack{j=1 \\ j\neq n-1}}^{n} a_{n-1,j} x_j}{a_{n-1,n-1}}$$

$$x_2 = \frac{b_2 - \sum_{\substack{j=1 \\ j\neq 2}}^{n} a_{2j}\, x_j}{a_{22}}$$

$$x_n = \frac{b_n - \sum_{\substack{j=1 \\ j\neq n}}^{n} a_{nj} x_j}{a_{nn}}$$

Gauss-Seidel Method and Jacobi method,

Algorithm

General Form for any row 'i'

$$x_i = \frac{b_i - \sum_{\substack{j=1 \\ j \neq i}}^{n} a_{ij} x_j}{a_{ii}}, i = 1,2, \dots, n.$$

How or where can this equation be used?

Gauss-Seidel Method and Jacobi method,

Solve for the unknowns

Assume an initial guess for [X]

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{n\text{-}1} \\ x_n \end{bmatrix}$$

Use rewritten equations to solve for each value of x_i.

Important: (Difference between Gauss-Seidel Method and Jacobi method)

Remember, in Gauss-Seidel Method the most recent value of x_i is used in the remaining calculations of the **same/ current** iteration i.e., x_{i+1} and so on.

Whereas, Jacobi method use the values of [X] in the next iteration

Example

Solve the following system of equations using (a) the Jacobi method, (b) the Gauss Seidel iteration method.

$$\begin{bmatrix} 4 & 2 & 1 \\ -1 & 2 & 0 \\ 2 & 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 11 \\ 3 \\ 16 \end{bmatrix}$$

$$4x_1 + 2x_2 + x_3 = 11$$
$$-x_1 + 2x_2 = 3$$
$$2x_1 + x_2 + x_3 = 16$$

Reorganize into new form:

$$x_1 = \frac{11}{4} - \frac{1}{2}x_2 - \frac{1}{4}x_3$$

$$x_2 = \frac{3}{2} + \frac{1}{2}x_1 - 0 \times x_3$$

$$x_3 = 4 - \frac{1}{2}x_1 - \frac{1}{4}x_2$$

(a) Jacobi method: use initial guess

$$x_1^0 = x_2^0 = x_3^0 = 1,$$

Stop when max $|x_1^k - x_1^{k-1}, x_2^k - x_2^{k-1}, x_3^k - x_3^{k-1}| \leq 0.1$

<u>First iteration:</u>

$$x_1^1 = (11/4) - (1/2)x_2^0 - (1/4)x_3^0 = 2$$
$$x_2^1 = (3/2) + (1/2)x_1^0 = 2$$
$$x_3^1 = 4 - (1/2)x_1^0 - (1/4)x_2^0 = 13/4$$

Example (cont.)

Second iteration: use the iterated values $x_1^1=2$, $x_2^1=2$, $x_3^1=13/4$

$$x_1^2 = (11/4) - (1/2)x_2^1 - (1/4)x_3^1 = 15/16$$
$$x_2^2 = (3/2) + (1/2)x_1^1 = 5/2$$
$$x_3^2 = 4 - (1/2)\, x_1^1 - (1/4)x_2^1 = 5/2$$

Converging Process:

$$[1,1,1],\ \left[2,2,\frac{13}{4}\right],\ \left[\frac{15}{16},\frac{5}{2},\frac{5}{2}\right],\ \left[\frac{7}{8},\frac{63}{32},\frac{93}{32}\right],\ \left[\frac{133}{128},\frac{31}{16},\frac{393}{128}\right],\ \left[\frac{519}{512},\frac{517}{256},\frac{767}{256}\right].$$

Stop the iteration when $\max\left|x_1^5 - x_1^4,\ x_2^5 - x_2^4,\ x_3^5 - x_3^4\right| \leq 0.1$

Final solution [1.014, 2.02, 2.996]

Exact solution [1, 2, 3]

Example (cont.)

(b) Gauss-Seidel iteration: Substitute the iterated values into the iterative process immediately after they are computed.

Use initial guess $X^0 = Y^0 = Z^0 = 1$

$$X = \frac{11}{4} - \frac{1}{2}Y - \frac{1}{4}Z,\ Y = \frac{3}{2} + \frac{1}{2}X,\ Z = 4 - \frac{1}{2}X - \frac{1}{4}Y$$

First iteration: $X^1 = \frac{11}{4} - \frac{1}{2}(Y^0) - \frac{1}{4}(Z^0) = 2$ Immediate substitution

$$Y^1 = \frac{3}{2} + \frac{1}{2}X^1 = \frac{3}{2} + \frac{1}{2}(2) = \frac{5}{2}$$

$$Z^1 = 4 - \frac{1}{2}X^1 - \frac{1}{4}Y^1 = 4 - \frac{1}{2}(2) - \frac{1}{4}\left(\frac{5}{2}\right) = \frac{19}{8}$$

Converging process: [1,1,1], $\left[2, \frac{5}{2}, \frac{19}{8}\right], \left[\frac{29}{32}, \frac{125}{64}, \frac{783}{256}\right], \left[\frac{1033}{1024}, \frac{4095}{2048}, \frac{24541}{8192}\right]$

The iterated solution [1.009, 1.9995, 2.996] and it converges faster

Assignment:

Given $x_1 = -2$, $x_2 = -1$, $x_3 = 1$ and $x_4 = 2$

1. Make a 4x4 |A| and a 4x1 |b| matrix like below
2. Solve the following system of equations using (a) the Jacobi method, (b) the Gauss Seidel iteration method.
3. Start with $x_1 = x_2 = x_3 = x_4 = 1$, and show that the values x like above.

$$\begin{vmatrix} 12 & -3 & 6 & 4 \\ 2 & 11 & 3 & 5 \\ 4 & 1 & 10 & 2 \\ -5 & 3 & 4 & -10 \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} -7 \\ -2 \\ 5 \\ -9 \end{vmatrix}$$